

MATTHEW A. COTAJ

Stamford, CT | (203) 541-1448 | cotajmatthew@gmail.com

EDUCATION

Georgetown University, Washington, DC

May 2026

Bachelor of Science in Chemistry (Honors), Minor in Biology | GPA: 3.46/4.00

Senior Honors Thesis in Chemistry (Advisor: Prof. Sarah L. Stoll): colloidal synthesis and magnetic characterization of lanthanide chalcogenide nanorods

Selected Coursework: Graduate-Level Organic Synthesis; Advanced Materials Chemistry (Nanomaterials, Polymers, Biomaterials, Quantum Materials); Fabrication & Characterization of Nanomaterials; Physical Chemistry I & II; Physical–Chemical Measurements; Inorganic Chemistry; Biochemistry; Mathematical Statistics

University of New South Wales (UNSW), Sydney, Australia

Jan 2025 – May 2025

Study Abroad – Coursework: Materials Chemistry, Bioinformatics, Physics, International Business

HONORS & AWARDS

- First Honors, Georgetown University (Fall 2025)
- Dean’s List, Georgetown University (Fall 2024, Spring 2026)
- Government of Albania Appointee: Youth NATO Summit Delegate (2024)

PUBLICATIONS

Hartman, T.; **Cotaj, M.**; El Jamal, E.S.; Stoll, S.L.* “Magnetic and magneto-optical properties of mixed cation and mixed valent EuLnCuQ₃ (Q = S, Se) nanorods.” *Manuscript in preparation.*

RESEARCH EXPERIENCE

Pfizer, Groton, CT

June 2026 – Present

Biopharmaceutics Intern, Drug Product Design (Pfizer Futures Program)

- Characterized V1/V2 biorelevant media across 20+ conditions via dynamic light scattering (DLS) and tensiometry, standardizing media preparation across four functions
- Correlated colloidal media properties to solubility and dissolution of five model compounds, improving in vitro–in vivo translation by ~30%
- Identified surfactant–bile micelle interactions forming solubility-limiting complexes, flagging excipient risks in early-phase formulations

Stoll Research Group, Georgetown University, Washington, DC

Feb 2023 – May 2026

Honors Thesis Researcher & Materials Science Undergraduate Research Fellow

- Achieved the first colloidal synthesis and magnetic characterization of phase-pure EuGdCuS₃ nanorods using single-source dithiocarbamate precursors under air-free conditions (Schlenk line, glovebox)
- Demonstrated compositional control of magnetism by substituting Gd³⁺ for Eu²⁺, raising saturation magnetization from 47.8 to ~78 emu/g (SQUID magnetometry)
- Screened lanthanide halide and sulfur precursors to phase-control binary (Nd₃S₄, Gd₃S₄) and ternary nanomaterials, verifying composition and morphology by PXRD, ICP-MS, EDS, and HRTEM
- Analyzed f-electron configurations and ligand-field effects to interpret nanoscale magnetic and electronic phenomena in lanthanide systems
- Presented honors thesis findings to the Department of Chemistry as the capstone of a three-year independent research program

Henkel, Stamford, CT

May 2025 – Aug 2025

Product Development Intern

- Developed and validated a method to differentiate liquid soap formulations, driving data-backed innovation for a \$440M product line
- Optimized formulation variables through data analysis, improving foam performance by 23% across all Dial products
- Supported antibacterial and non-antibacterial soap development under FDA and OTC compliance

CG Pharmaceuticals, Inc., Washington, DC

Mar 2024 – July 2024

Rare Disease Therapeutics R&D Intern

- Researched alternative treatments for neurodegenerative disorders (Friedreich’s Ataxia, Fragile X Syndrome), evaluating drug targets and biological pathways for feasibility based on IP ownership and clinical relevance

- Conducted scientific and intellectual-property research supporting seed-stage biotech strategy

PRESENTATIONS

- Senior Honors Thesis Presentation, Department of Chemistry, Georgetown University, Washington, DC (Spring 2026)
- “Formulation Strategies and Theoretical Basis for a Next-Generation Chassis Designed to Improve Foaming Performance in Dial and Henkel Systems,” Henkel Executive R&D Board Meeting, Trumbull, CT (Aug 2025)
- Weekly research talks on nanorod synthesis and characterization, Stoll Research Group, Georgetown University (Jan 2024 – May 2026)

TEACHING & OUTREACH

Georgetown University, Department of Chemistry **Aug 2025 – May 2026**
General Chemistry I & II Laboratory Teaching Assistant

- Guided students through theoretical and wet-chemistry experiments, reinforced safe and accurate laboratory technique, and maintained equipment and reagents across weekly lab sections

GUMSHOE (Georgetown University Math & Science Hands-On Enrichment) **Sept 2023 – May 2026**
Math & Science Tutor

- Provided hands-on tutoring in math, chemistry, biology, and physics to K–12 students across Washington, DC

Dynamite Debate, Kids Helping Kids **Sept 2021 – Jan 2025**
Founder & Instructor

- Founded and taught a free debate-education program for K–12 students from underserved communities, delivering multi-month courses in multiple competitive styles

Varsity Tutors **June 2022 – Aug 2023**
Student Tutor – math through pre-calculus, chemistry, biology, physics, and SAT/ACT preparation

TECHNICAL SKILLS

Synthesis & Fabrication: Air-free synthesis (Schlenk line, glovebox); colloidal nanomaterial synthesis; chemical vapor deposition (CVD); sputtering; thin-film deposition; photolithography; etching

Characterization: PXRD (Ultima IV); SQUID magnetometry; TEM/HRTEM; SEM; XPS; EDX/EDS; AFM; ICP-OES; ICP-MS; NMR; IR/FTIR; UV-Vis; Raman; DLS; tensiometry; cyclic voltammetry; surface plasmon resonance; HPLC/UPLC; LC-MS; GC-MS; RT-PCR; gel electrophoresis

Computational: Python; R; OriginLab; JMP; ChemDraw; Gaussian; Spartan; Cambridge Structural Database (CSD); MDI Jade; SpinWorks; MEGA; Snowflake

Safety Training: GxPs; HAZMAT; Highly Potent Compounds (HPCs); Laser Safety; Radioactive Materials (RAM)

ADDITIONAL EXPERIENCE

ANAMA Package & Container Testing Services, Stamford, CT **Summers 2022 – 2024**
Project Manager Intern | Package Testing Technician

- Developed and executed UN/USDOT/ISTA testing protocols for clients including Bloomberg, Duracell, and the U.S. Army; managed resources and budgets to double per-diem project completion

Georgetown University Alumni Association, Washington, DC **Aug 2024 – Sept 2025**
Operations & Marketing Associate

- Managed platform operations and user access, built alumni–student connections through events, and used data analytics to guide organizational decision-making

LEADERSHIP & SERVICE

- Head of Marketing, Georgetown University Albanian Society (Sept 2023 – May 2026)
- Project Leader & Senior Ambassador, Kids Helping Kids (June 2016 – Present)
- Treasurer, Georgetown University Chemistry Club (Sept 2024 – Aug 2025)

MEMBERSHIPS & LANGUAGES

Memberships: American Chemical Society (2024 – Present); Albanian Academic and Professional Association (2023 – Present)

Languages: Spanish (spoken, written, and comprehension)